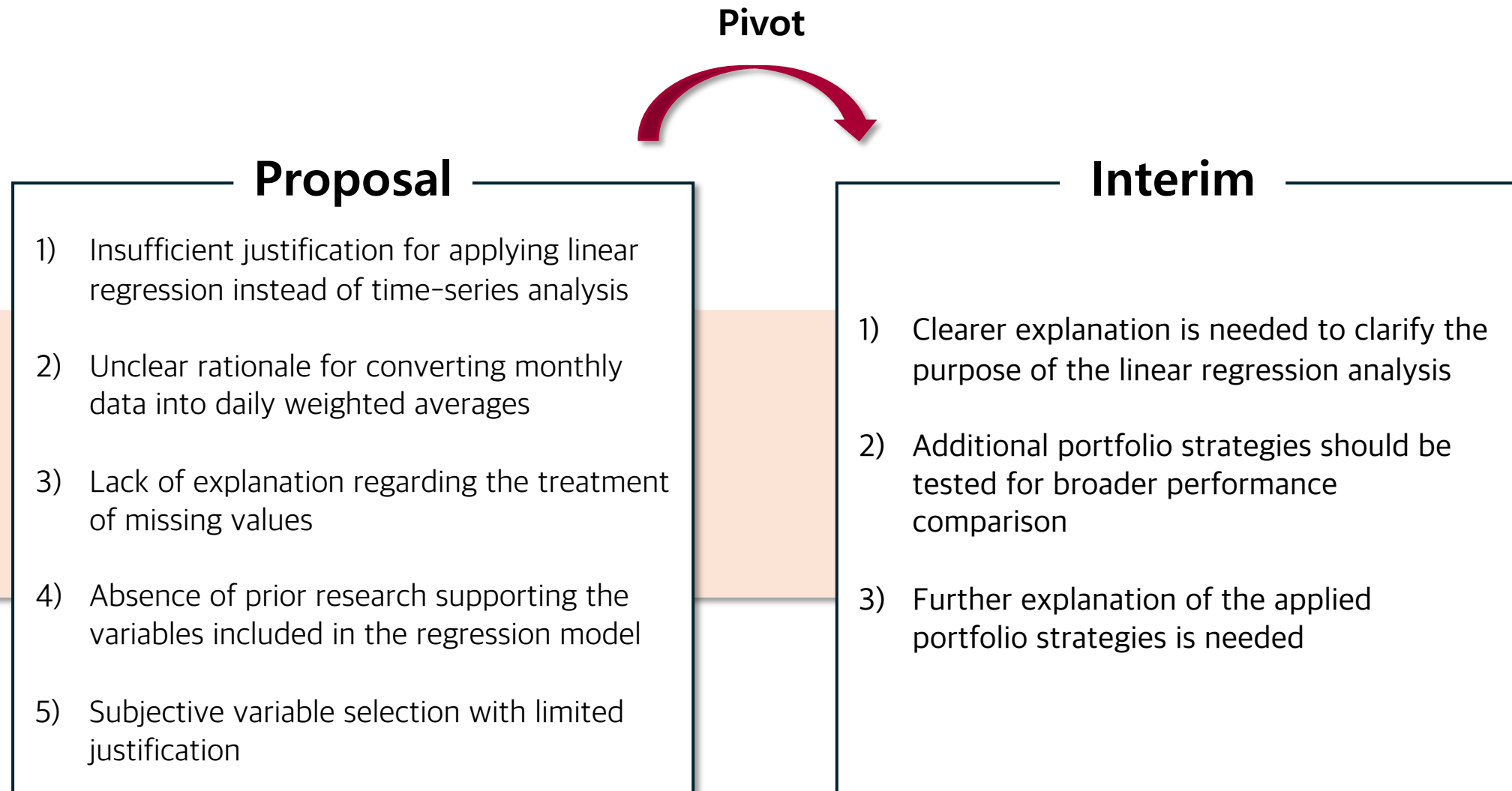


2025-Winter

Are Sector ETF Strategies *Alpha-Generating* or *Factor-Driven*?

Equal-Weight vs Max-Sharpe vs Min-Variance

2021002297 BSBA SangWoo Chon



B E F O R E



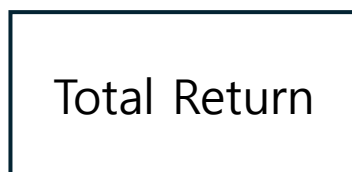
Identifying Key Factors Influencing the Korean Won Exchange Rate

" P I V O T "

Reason#1: research topic unsuitable for simple linear regression

Reason#2: Methodological flaws in converting monthly data into daily weighted averages

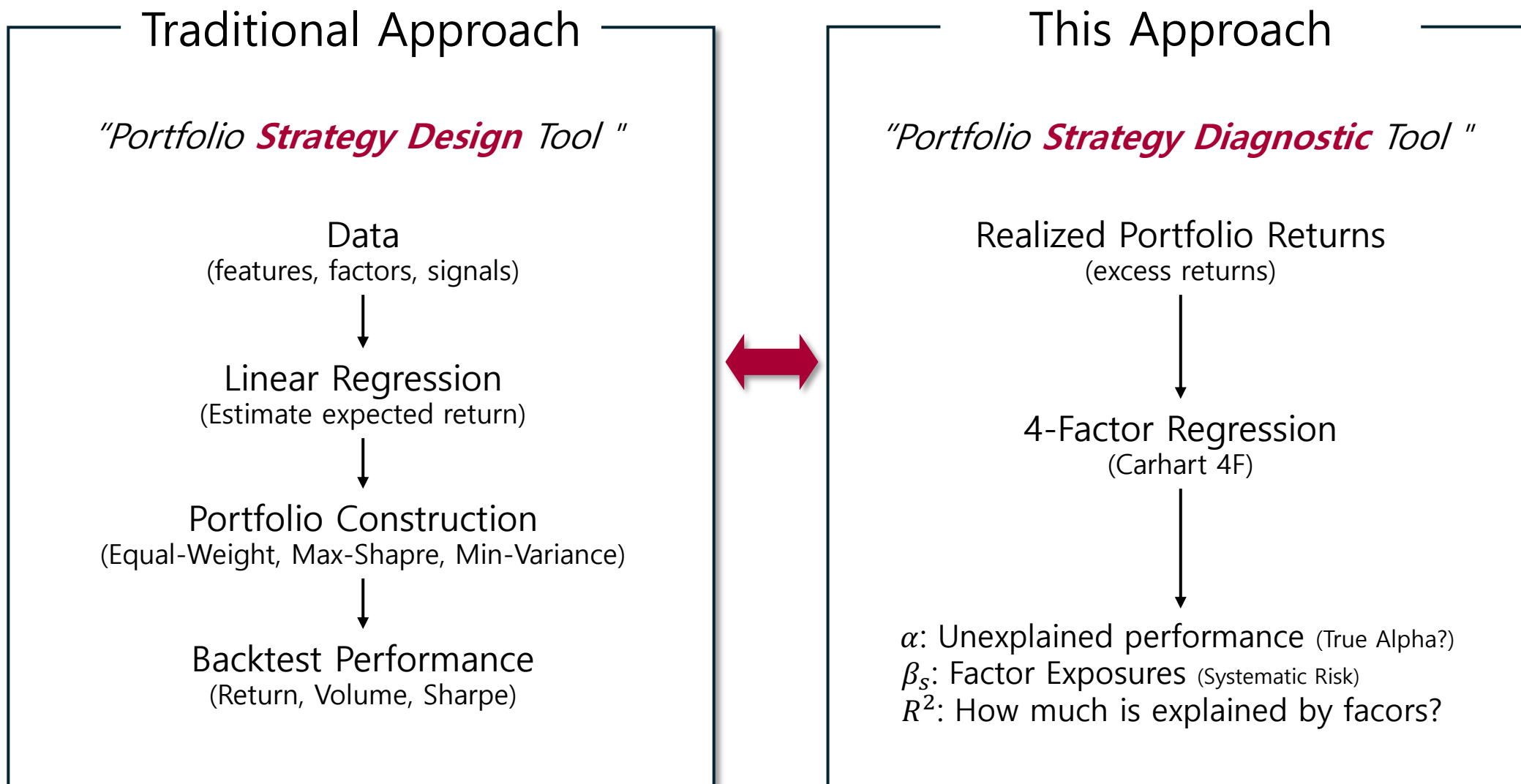
A F T E R



Performance driven by exposure to specific factor risks (Beta)

Strategy-specific excess returns (Alpha)

Decomposes portfolio excess returns into strategy-specific skill (alpha) and factor exposure (beta)



Goal

- ① Estimating **Factor Exposures (β)** and **Risk-Adjusted Performance (α)** of Sector ETF Portfolios Using Regression Analysis
- ② Assess whether portfolio outperformance reflects **strategy-specific skill** or **compensation** for factor exposure



1

Equal-Weight

2

Max-Sharpe

3

Min-Variance



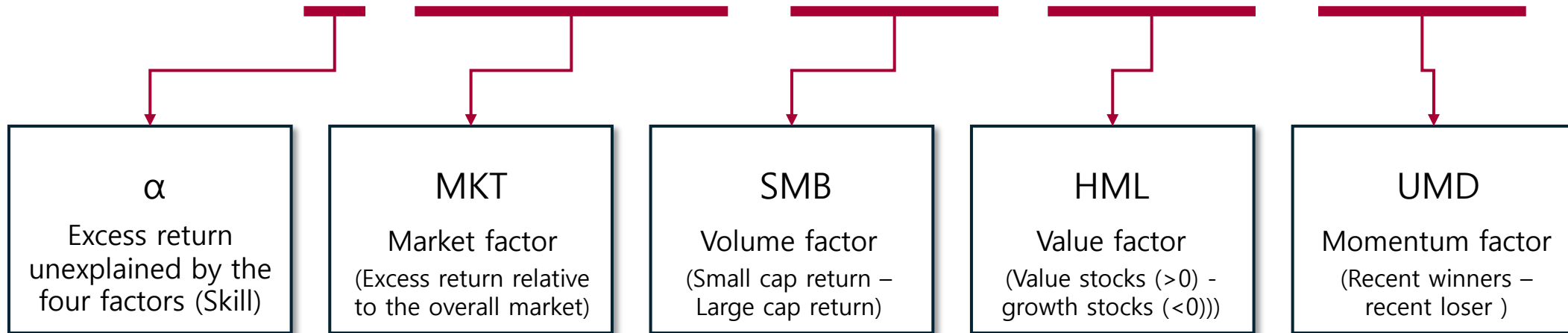
Apply the Carhart four-factor regression to each portfolio strategy to achieve Objective ①

Using regression results from Objective ①, determine Objective ②.

Carhart 4-Factor Model

The Carhart 4-factor model extends the Fama–French 3-factor model by adding momentum (UMD), allowing it to control for trend-following and rebalancing effects commonly observed in portfolio optimization strategies and more accurately evaluate risk-adjusted performance (α).

$$R_t^{excess} = \alpha + \beta_M Mkt_t^{excess} + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t$$



Equal-Weight

Equal weight across all sector ETFs

Max-Sharpe

Optimized Investment Strategy
(Maximizing Risk-Adjusted Returns)

Asset allocation approach based on William Sharpe's Sharpe ratio that identifies the portfolio with the highest excess return per unit of volatility

$$E(R_p) = w^T \mu, \sigma_p = \sqrt{w^T \Sigma w} (\text{Variance})$$

$$\text{Sharpe} = \frac{w^T (\mu - r_f \mathbf{1})}{\sqrt{w^T \Sigma w}}$$

Let (w) denote the vector of asset weights. The portfolio's expected return, volatility, and Sharpe ratio are defined as shown above.

Min-Variance

Defensive Investment Strategy

Strategy that minimizes overall portfolio volatility to reduce downside losses and pursue stable returns.

Given (N) sector ETFs and covariance matrix (Σ), the minimum-variance portfolio is defined as:

$$\min_w \sigma_p^2 = w^T \Sigma w$$

$$\mathbf{1}^T w = 1, w^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

$w^T \Sigma w$ (variance) Minimize portfolio variance while ensuring that the investment weights sum to 1.

AQR Data

QMJ Factors

MKT, SMB,

HML, UMD

1927 ~ 2025.10

Yahoo Fin.

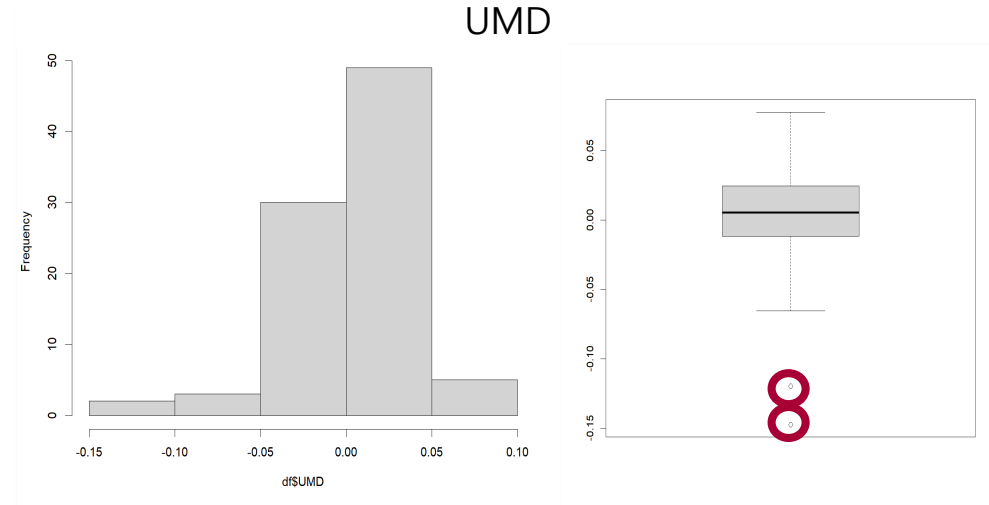
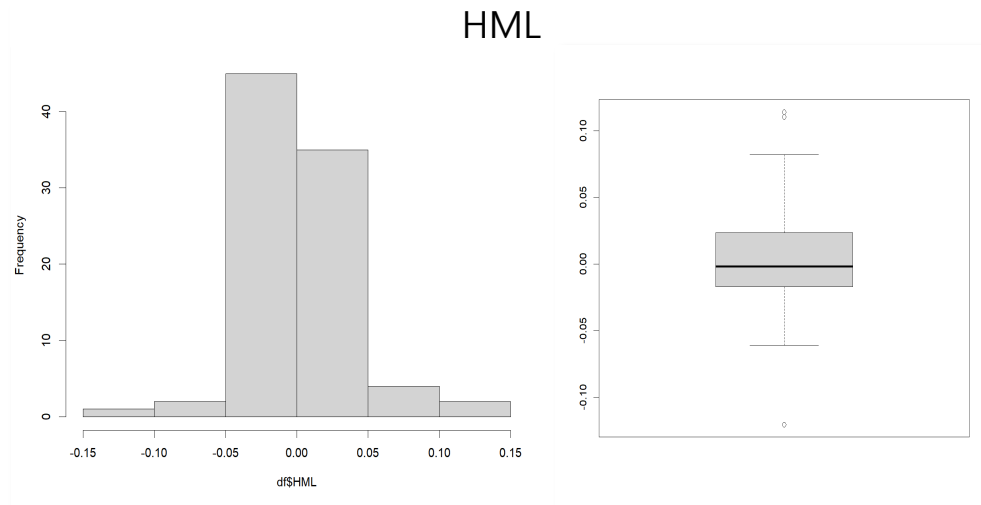
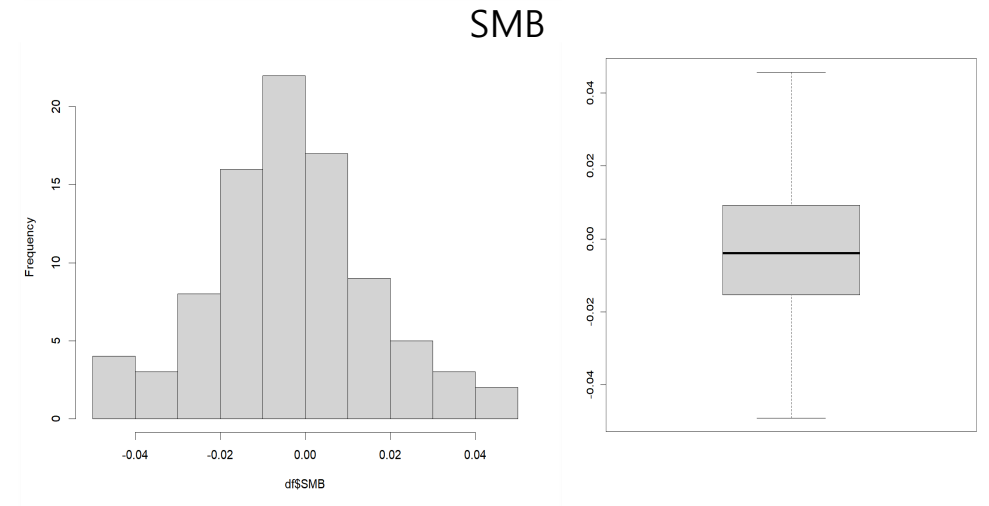
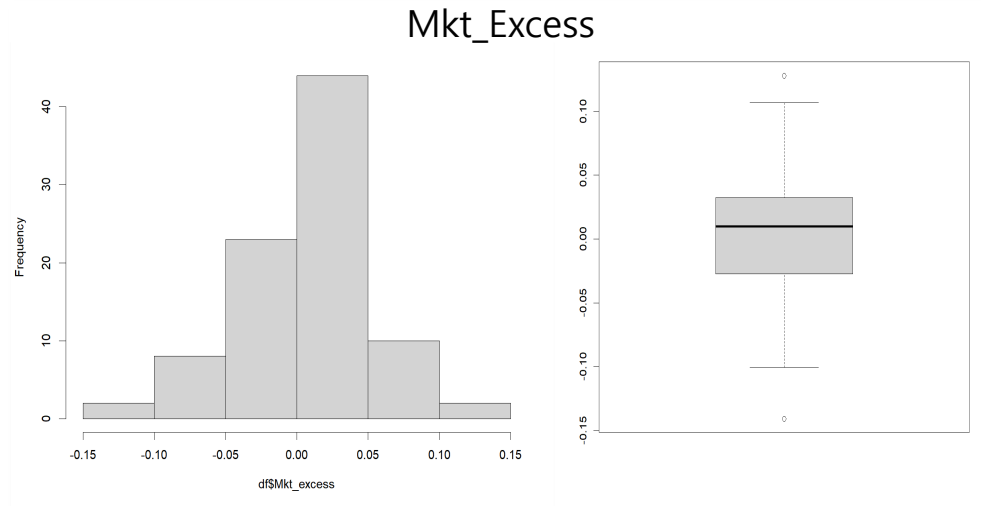
11 Sector ETF

Price

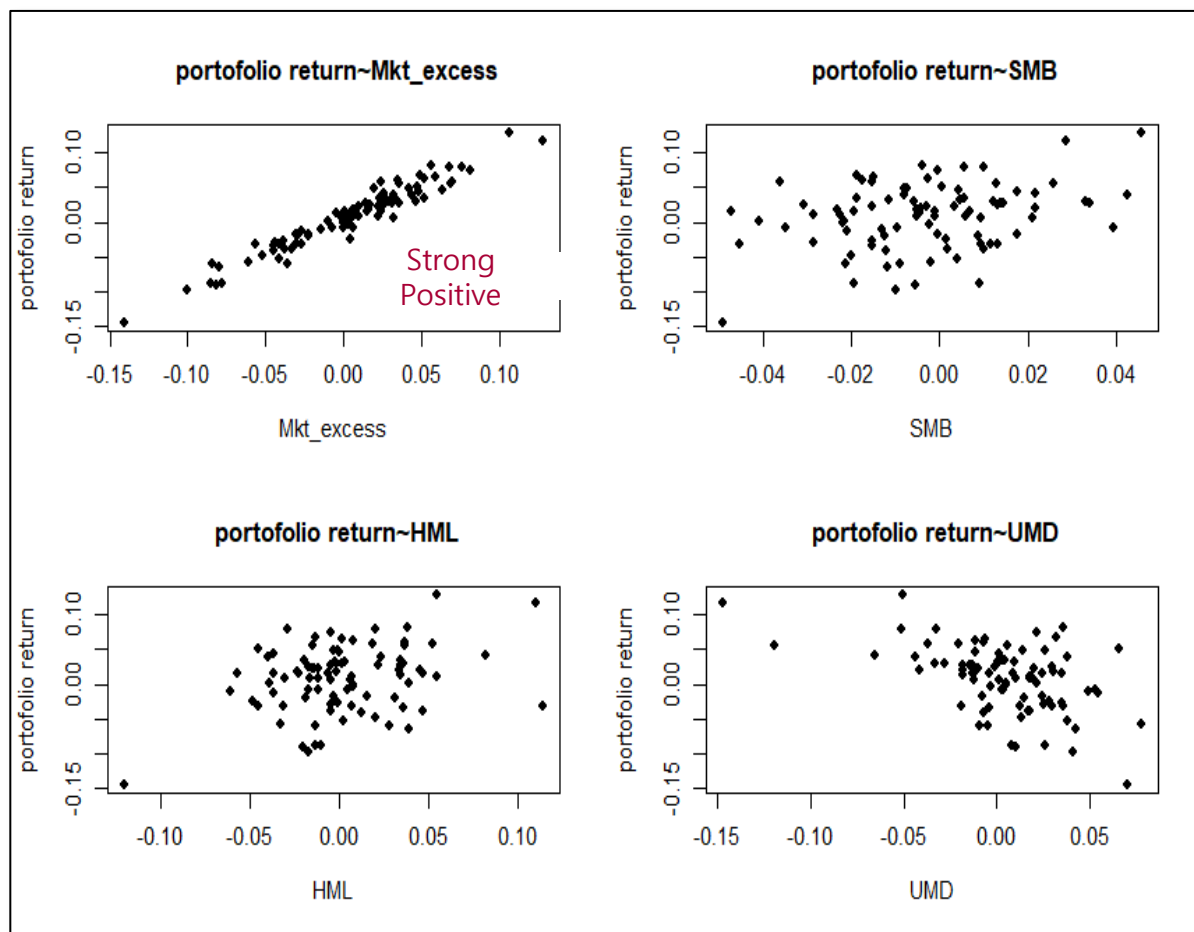
2018.07 ~

Variable Type	Notation	Definition	Missing Values
Dependent	R_excess	Portfolio Return – Risk Free Rate (Equal – Weight)	0
Independent	Mkt_excess	Market – Risk Free Rate	0
Independent	SMB	Small – Big (Size Factor)	0
Independent	HML	High – Low (Value Factor)	0
Independent	UMD	Up – Down (Momentum Factor)	0

✓ **Data Integrity Check:** No missing values; all variables contain complete monthly time-series data



➤ Overall distributions are stable, with only a few potential outliers observed in UMD



- ✓ Equal-weighted sector ETF portfolio is broadly diversified across the market and is therefore likely to have **substantial exposure to the market factor** (Mkt_excess)
- ✓ Since size, value, and momentum factors reflect firm-specific investment styles, the equal-weighted sector portfolio is **hypothesized to remain relatively neutral to SMB, HML, and UMD**
- ✓ Figure provides partial support for this intuition

$$R_t^{excess} = \alpha + \beta_M Mkt_t^{excess} + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t$$

<Figure 1> Factor Variance Correlation

SMB	1.00	0.20	-0.41	0.39
HML	0.20	1.00	-0.62	0.24
UMD	-0.41	-0.62	1.00	-0.53
Mkt Excess	0.39	0.24	-0.53	1.00
	SMB	HML	UMD	Mkt Excess

[Table 1] VIF

Variable	VIF
SMB	2.3032
HML	1.6582
UMD	1.2754
Mkt_Excess	1.4858

➤ All 'VIF < 5', no multicollinearity

Regression Models

* Only the maximum Sharpe ratio strategy exhibits significant alpha excess returns unexplained by the four factors.

$$R_t^{excess} = \alpha + \beta_M Mkt_t^{excess} + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t$$

[Table 2] Sector Equal-Weight Portfolio

[Table 3] Max Sharpe Ratio Portfolio

[Table 4] Min Variance Portfolio

Variable	Estimate	Std.Error r	t-value	Pr(> t)	Variable	Estimate	Std.Error r	t-value	Pr(> t)	Variable	Estimate	Std.Error r	t-value	Pr(> t)
(Intercept)	0.0015	0.0012	1.295	0.1988	(Intercept)	0.0099	0.0023	4.296	P < 0.001	(Intercept)	0.0015	0.0022	1.295	0.6846
Mkt_excess	1.0179	0.0299	34.031	P < 0.001	Mkt_excess	1.0972	0.0573	19.134	P < 0.001	Mkt_excess	0.7532	0.0549	34.031	P < 0.001
SMB	-0.1767	0.0665	-2.656	P < 0.01	SMB	-0.3115	0.1275	-2.442	P < 0.05	SMB	-0.5180	0.1223	-2.656	P < 0.001
HML	0.1808	0.0416	4.346	P < 0.001	HML	-0.5684	0.0797	-7.125	P < 0.001	HML	0.1576	0.0764	4.346	P < 0.05
UMD	0.1063	0.0509	2.087	P < 0.05	UMD	0.1808	0.0976	-1.853	P < 0.1	UMD	0.2098	0.0936	2.087	P < 0.05

Residual Standard Error: 0.01076

R-squared: 0.9508 Adj. R-squared: 0.9485

p-value: < 0.001

Residual Standard Error: 0.02062

R-squared: 0.8638 Adj. R-squared: 0.8573

p-value: < 0.001

Residual Standard Error: 0.01977

R-squared: 0.7196 Adj. R-squared: 0.7063

p-value: < 0.001

Equal-Weight

$$R_t^{excess} = 0.001 + 1.017 * Mkt_t^{excess} - 0.176 * SMB_t + 0.180 * HML_t + 0.106 * UMD_t$$

Residual Standard Error:

0.0107

R-squared: 0.950

Adj. R-squared: 0.948

: With no significant alpha and a (Mkt_excess) beta close to 1, the strategy closely tracks the market, suggesting that simple diversification alone is insufficient to outperform on a risk-adjusted basis.

Max-Sharpe

$$R_t^{excess} = 0.009 + 1.097 * Mkt_t^{excess} - 0.311 * SMB_t - 0.568 * HML_t + 0.180 * UMD_t$$

Residual Standard Error: 0.0206

R-squared: 0.863

Adj. R-squared: 0.857

: Consistent with its objective of maximizing returns for a given level of risk, the strategy exhibits a more aggressive market profile and generates significant alpha unexplained by factor exposures $\alpha + \beta > 1$

Min-Variance

$$R_t^{excess} = 0.001 + 0.753 * Mkt_t^{excess} - 0.518 * SMB_t + 0.157 * HML_t + 0.209 * UMD_t$$

Residual Standard Error: 0.0197

R-squared: 0.719

Adj. R-squared: 0.706

: Consistent with its defensive objective of reducing volatility during market downturns, the strategy showed lower market sensitivity than the others and a stronger large-cap tilt relative to volatile small-cap stocks.